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PROJECT NOTES: AUTOMATIC TRAFFIC LIGHT

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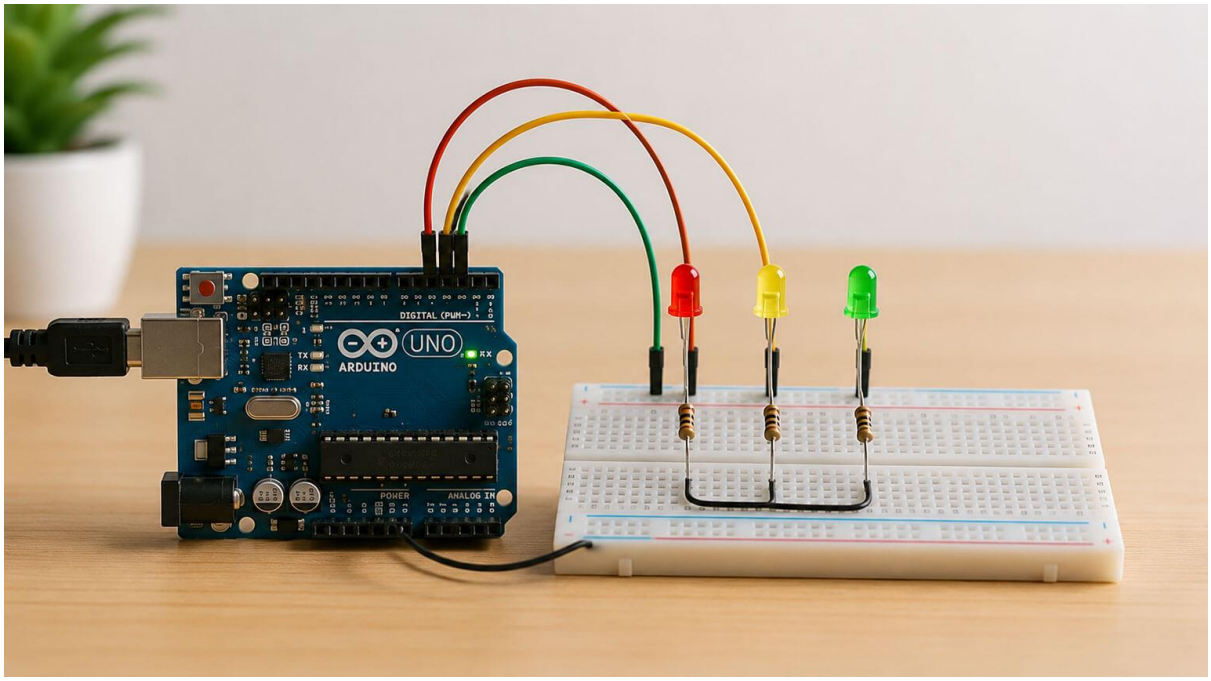
1. PROJECT OVERVIEW

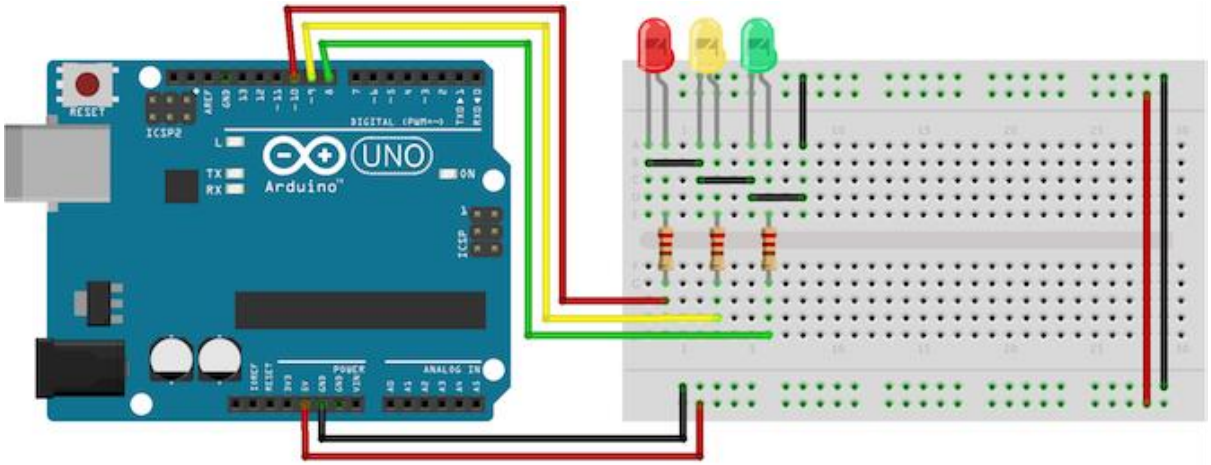
This project creates an automatic traffic light system. It uses an Arduino Uno to control three LEDs (Red, Yellow, Green) in a timed loop. This mimics a real street intersection signal.

2. COMPONENTS REQUIRED & THEIR USES

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* Arduino Uno: The microcontroller board. It acts as the "brain" of the circuit, executing code to control electricity flow.
* 3 LEDs (Red, Yellow, Green): Light Emitting Diodes. They represent the traffic signals (Stop, Ready, Go).
* 3 Resistors (220 ohms): Current-limiting components. They protect the LEDs from drawing too much current and burning out.
* Breadboard: A solderless grid board. It allows easy connection of components and jumper wires.
* Jumper Wires: Conducting wires used to bridge electrical connections between the Arduino and the breadboard.





3. HARDWARE STEP-BY-STEP GUIDELINES

Follow these steps to assemble the circuit on the breadboard:

[Step 1] Place the LEDs

* Insert the Red, Yellow, and Green LEDs onto separate rows of the breadboard.

* Note: Each LED has a longer leg (Anode/Positive +) and a shorter leg (Cathode/Negative -).

[Step 2] Connect the Ground Rail

* Run a jumper wire from the "GND" pin on the Arduino to the negative (-) rail (the long side strip) of the breadboard.

[Step 3] Add Resistors

* For each LED, connect a 220-ohm resistor from its shorter leg (Cathode) to the negative (-) rail of the breadboard. This provides a clear ground path for all 3 lights.

[Step 4] Connect Digital Pins

* Connect a jumper wire from the longer leg (Anode) of the Red LED to Digital Pin 10 on the Arduino.
* Connect a jumper wire from the longer leg (Anode) of the Yellow LED to Digital Pin 9 on the Arduino.
* Connect a jumper wire from the longer leg (Anode) of the Green LED to Digital Pin 8 on the Arduino.

4. ARDUINO PROGRAM CODE

Type this exact code into your Arduino IDE and upload it to your board:

```
// Define the digital pins for each LED
int redLed = 10;
int yellowLed = 9;
int greenLed = 8;

void setup() {
  // Set all LED pins as outputs
  pinMode(redLed, OUTPUT);
  pinMode(yellowLed, OUTPUT);
  pinMode(greenLed, OUTPUT);
```

```
}

void loop() {
  // 1. Red Light ON (Green and Yellow OFF) - Cars Stop
  digitalWrite(redLed, HIGH);
  digitalWrite(yellowLed, LOW);
  digitalWrite(greenLed, LOW);
  delay(5000); // Wait 5 seconds

  // 2. Green Light ON (Red and Yellow OFF) - Cars Go
  digitalWrite(redLed, LOW);
  digitalWrite(yellowLed, LOW);
  digitalWrite(greenLed, HIGH);
  delay(5000); // Wait 5 seconds

  // 3. Yellow Light ON (Red and Green OFF) - Cars Slow Down
  digitalWrite(redLed, LOW);
  digitalWrite(yellowLed, HIGH);
  digitalWrite(greenLed, LOW);
  delay(2000); // Wait 2 seconds
}
```

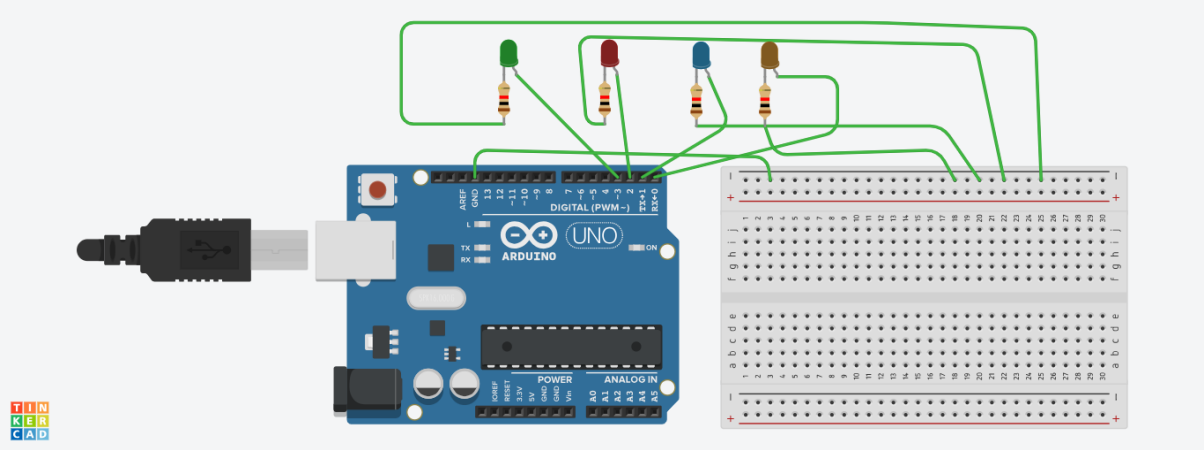
5. OPERATIONAL SEQUENCE SUMMARY

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* Stage 1: Red LED lights up for 5000 milliseconds (5 seconds).
* Stage 2: Red turns off, Green LED lights up for 5000 milliseconds (5 seconds).
* Stage 3: Green turns off, Yellow LED lights up for 2000 milliseconds (2 seconds).
* The cycle automatically repeats indefinitely.

2. Disco Light show

https://www.tinkercad.com/things/8HfE7d2rWd4/editel?returnTo=%2Fdashboard&sharecode=KakpEXxI8F45rER40wcINxvI3fELPat6DGZS06KLvGc



Disco Light Show using Tinkercad Circuits

This is a beginner-friendly Arduino project for students. The LEDs blink in a repeated sequence to create a disco light effect.

1. Components required



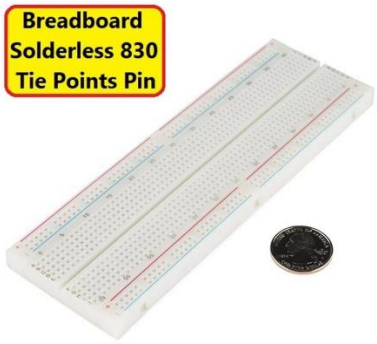
Arduino Uno R3

1 board — controls the LEDs.



LEDs

4 LEDs — red, yellow, green and blue.



Breadboard

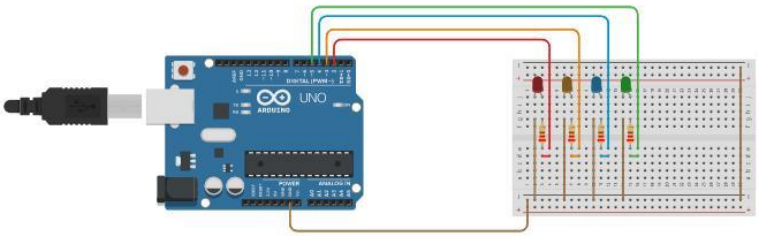
1 — for connecting the circuit.



220 Ω resistors and jumper wires

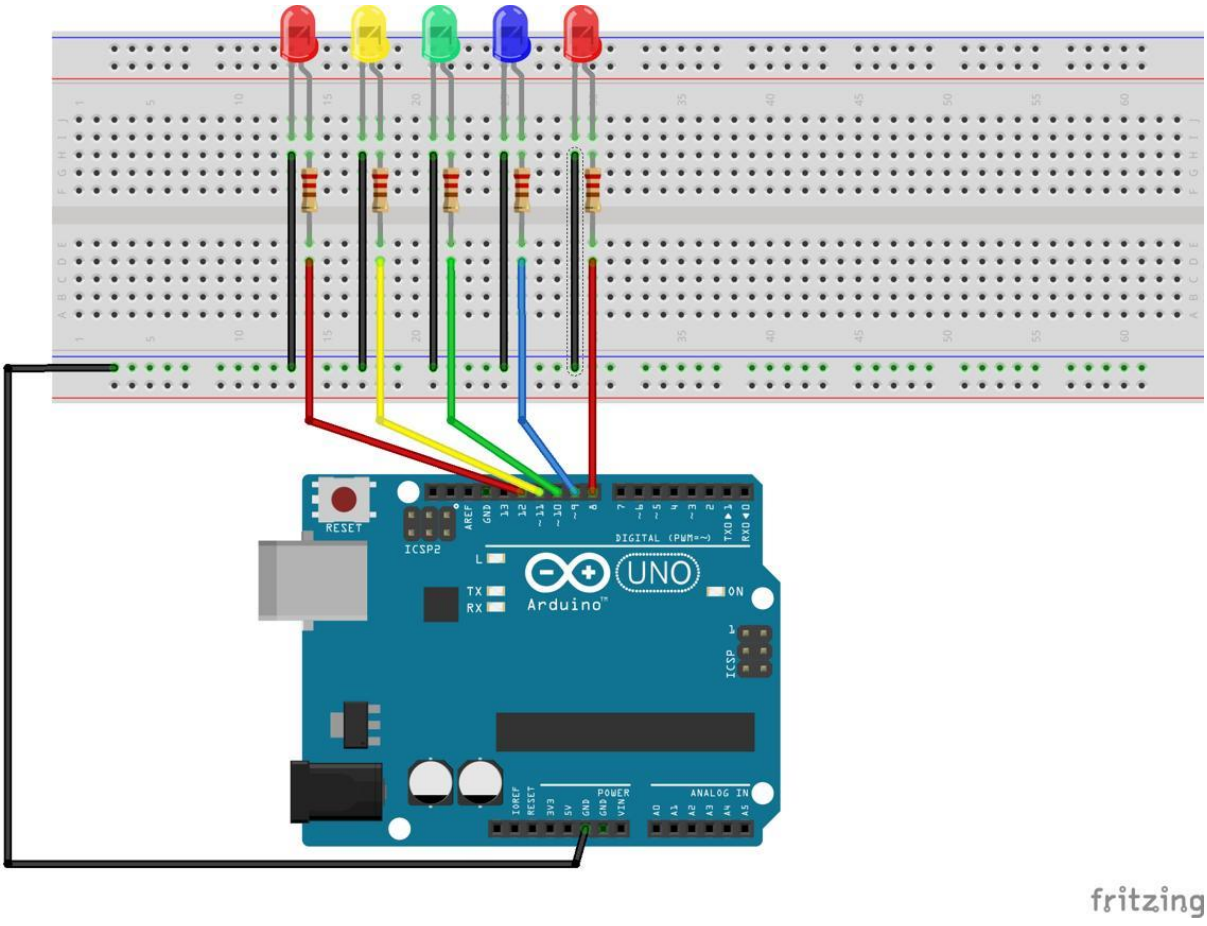
One resistor for each LED.

2. Create the circuit in Tinkercad



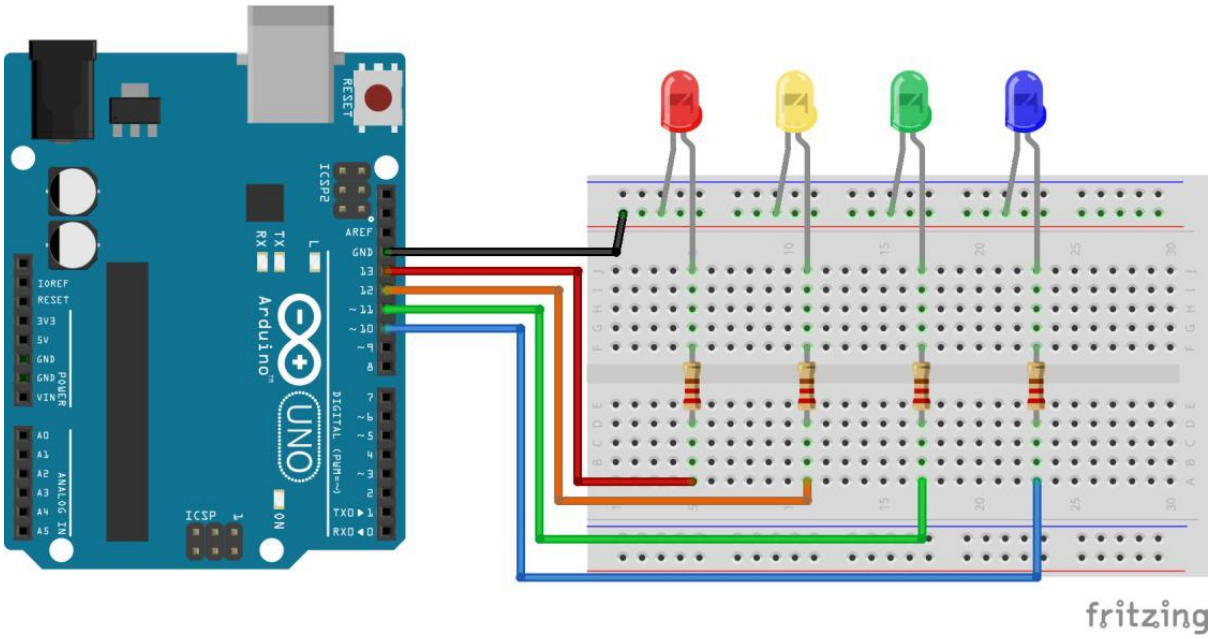
Step 1: Open Tinkercad

1. Go to Tinkercad.
2. Sign in and open Circuits.
3. Click Create new Circuit.



Step 2: Add components

Drag an Arduino Uno, breadboard, four LEDs, four 220 Ω resistors and jumper wires into the workspace.



Step 3: Connect the LEDs

Connect the LEDs to digital pins 8, 9, 10 and 11.

- Each Arduino pin connects through a 220 Ω resistor to an LED's anode (long leg).
- Connect each LED's cathode (short leg) to GND.

3. Program using CodeBlocks

In Tinkercad, click Code → Blocks. Create the following logic using the blocks available in your editor.

When the program starts

Set pin 8 to HIGH

Wait 0.2 seconds

Set pin 8 to LOW

Set pin 9 to HIGH

Wait 0.2 seconds

Set pin 9 to LOW

Set pin 10 to HIGH

Wait 0.2 seconds

Set pin 10 to LOW

Set pin 11 to HIGH

Wait 0.2 seconds

Set pin 11 to LOW

Repeat forever

Place the four pin operations inside a forever/repeat loop so the light pattern continues.

4. Alternative: Arduino code

If you switch the Code menu to Text, you can use this program:

```
int leds[] = {8, 9, 10, 11};

void setup() {
  for (int i = 0; i < 4; i++) {
    pinMode(leds[i], OUTPUT);
  }
}

void loop() {
  // Lights move forward
  for (int i = 0; i < 4; i++) {
    digitalWrite(leds[i], HIGH);
    delay(200);
    digitalWrite(leds[i], LOW);
  }

  // Lights move backward
  for (int i = 2; i >= 1; i--) {
    digitalWrite(leds[i], HIGH);
    delay(200);
    digitalWrite(leds[i], LOW);
  }
}
```

5. Run and observe

1. Click Start Simulation.
2. Watch the four LEDs light up one after another.
3. The sequence repeats to create a disco-style moving light effect.
4. Change `delay(200)` to `delay(100)` for faster lights or `delay(500)` for slower lights.

Expected result: The LEDs blink in a repeating forward-and-backward pattern.

Classroom explanation: “The Arduino switches each LED ON and OFF using digital pins. The `delay()` function controls how long each LED stays ON, creating the disco light effect.”